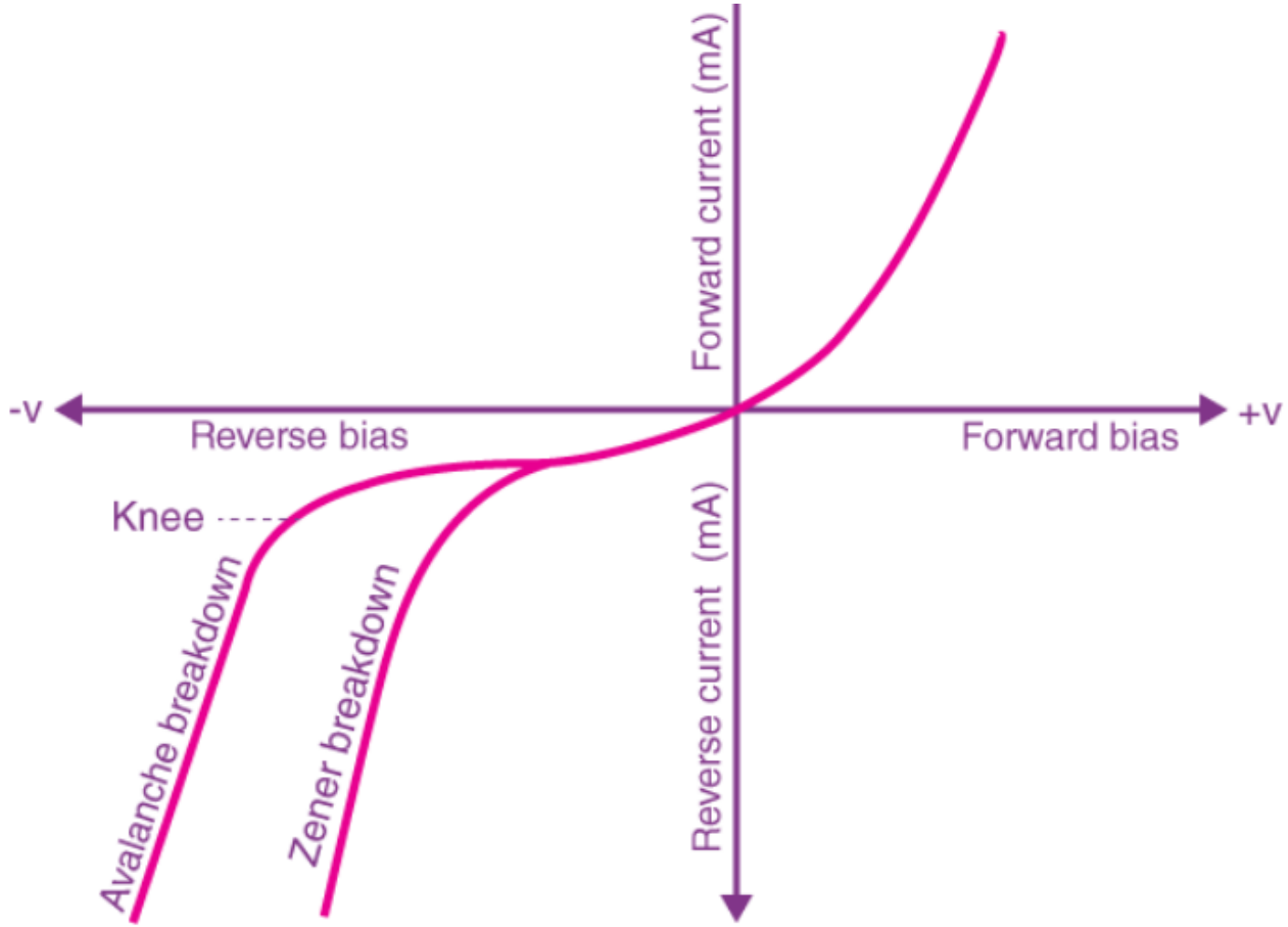
Zener Breakdown

- When the diode is reverse biased, the kinetic energy of the electrons increases, and they move at a high velocity.
- The high-velocity electrons collide with other atoms and give rise to free electrons.
- These free electrons, in turn, give rise to a high value of reverse saturation current.
- This is known as Zener breakdown.

Avalanche Breakdown

- The avalanche breakdown occurs when a high reverse voltage is applied across the diode.
- As we increase the applied reverse voltage, the electric field across the junction increases.
- This electric field exerts a force on the electrons at the junction and frees them from covalent bonds.
- These free electrons start moving with high velocity across the junction and collide with the other atoms, thus creating more free electrons.
- This results in a rapid increase in net current.
- Both these breakdowns occur in Zener diodes.



Volt-Ampere Characteristics of Si and Ge Zener Diode

Zener Breakdown	Avalanche Breakdown
The process in which the electrons move across the barrier from the valence band of p-type material to the conduction band of n-type material is known as Zener breakdown.	The process of applying high voltage and increasing the free electrons or electric current in semiconductors and insulating materials is called an avalanche breakdown.
This is observed in Zener diodes having a Zener breakdown voltage V_z of 5 to 8 volts.	This is observed in Zener diode having a Zener breakdown voltage V_z greater than 8 volts.
The valence electrons are pulled into conduction due to the high electric field in the narrow depletion region.	The valence electrons are pushed to conduction due to the energy imparted by accelerated electrons, which gain their velocity due to their collision with other atoms.
The increase in temperature decreases the breakdown voltage.	The increase in temperature increases the breakdown voltage.
The VI characteristics of a Zener breakdown has a sharp curve.	The VI characteristic curve of the avalanche breakdown is not as sharp as the Zener breakdown.
It occurs in diodes that are highly doped.	It occurs in diodes that are lightly doped.